Project Report of
Applied Internet of Things (AIOT)
(CSE 2198)

AI-Based Stethoscope for Autonomous Healthcare Diagnostics and Medical Assessment.

Submitted by

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Abstract

The rapid advancement of artificial intelligence (AI) and embedded systems has paved the way for innovative solutions in modern healthcare. This project presents the development of an **AI-based stethoscope** designed for **autonomous diagnostics and medical assessment**, particularly in cardiopulmonary health. Traditional stethoscopes rely heavily on the clinician's auditory skill, often leading to misinterpretation and missed early signs of diseases. Our system enhances this process by integrating a digital microphone with a **Raspberry Pi Zero 2 W**, which captures auscultation sounds and processes them using **machine learning algorithms**.

The system converts heart and lung sounds into digital signals, applies **signal processing techniques** to extract meaningful features such as MFCCs and spectrograms, and utilizes a trained **convolutional neural network (CNN)** to classify sounds as normal or indicative of conditions like **murmurs, arrhythmias, wheezes, or crackles**. The device operates in real time and supports **wireless data transmission**, making it ideal for **remote diagnostics and telemedicine** applications.

By combining affordability, portability, and intelligent analysis, the AI-based stethoscope aims to improve **early disease detection**, **reduce misdiagnoses**, and extend quality healthcare to **underserved regions**. The prototype demonstrates the potential of AI-powered medical devices in transforming the future of primary diagnostics and personalized healthcare.

Contents

Chapter 01: Introduction

1.1. Introduction

The integration of Artificial Intelligence (AI) in medical devices is revolutionizing healthcare delivery, particularly in diagnostics. One such innovation is the AI-powered stethoscope, which augments traditional auscultation with intelligent sound analysis. This project explores the development of an AI-based stethoscope that enables real-time, automated detection of heart and lung anomalies, with the goal of enhancing telemedicine and remote diagnostics.

1.2. Background

Traditional stethoscopes rely on the clinician's auditory interpretation, which can vary in accuracy. Misdiagnoses often arise due to human error, lack of expertise, or background noise. The rise of digital health tools and low-cost computing platforms like the Raspberry Pi Zero 2 W makes it possible to embed machine learning models into compact, portable devices. By converting heart/lung sounds into digital signals and analyzing them via AI, diagnostic capabilities can be extended to underserved and remote areas.

1.3. Project Objectives

- Develop a smart stethoscope capable of capturing, digitizing, and analyzing auscultation sounds.

- Train and deploy AI models (e.g., CNNs) for classification of common cardiopulmonary abnormalities.

- Optimize system for real-time analysis on low-power hardware (Raspberry Pi Zero 2 W).

- Enable wireless data transfer and result visualization using a mobile or web interface.

- Support remote diagnostics and integration with telemedicine platforms.

1.4. Scope

The project encompasses:

- ❖ Hardware design: integrating a microphone, USB sound card, and Raspberry Pi.
- ❖ Signal processing: converting and filtering audio signals.
- ❖ Machine learning: training and deploying models for anomaly classification.

- ❖ User interface: presenting results via local output or cloud-based dashboards.
- ❖ Does not include clinical trials or FDA certification, but focuses on a functional prototype for research and demonstration.

1.5. Project Management

- **Team Structure:** Divided into hardware, software, and AI development teams.

- **Development Timeline:**

  - ❖ Week 1–2: Component selection and circuit design.
  - ❖ Week 3–5: Audio data collection and preprocessing.
  - ❖ Week 6–8: Model training and testing.
  - ❖ Week 9–10: Integration and testing on Raspberry Pi.

- **Tools Used:** GitHub (version control), Trello (task management), Python (development), TensorFlow/PyTorch (ML).

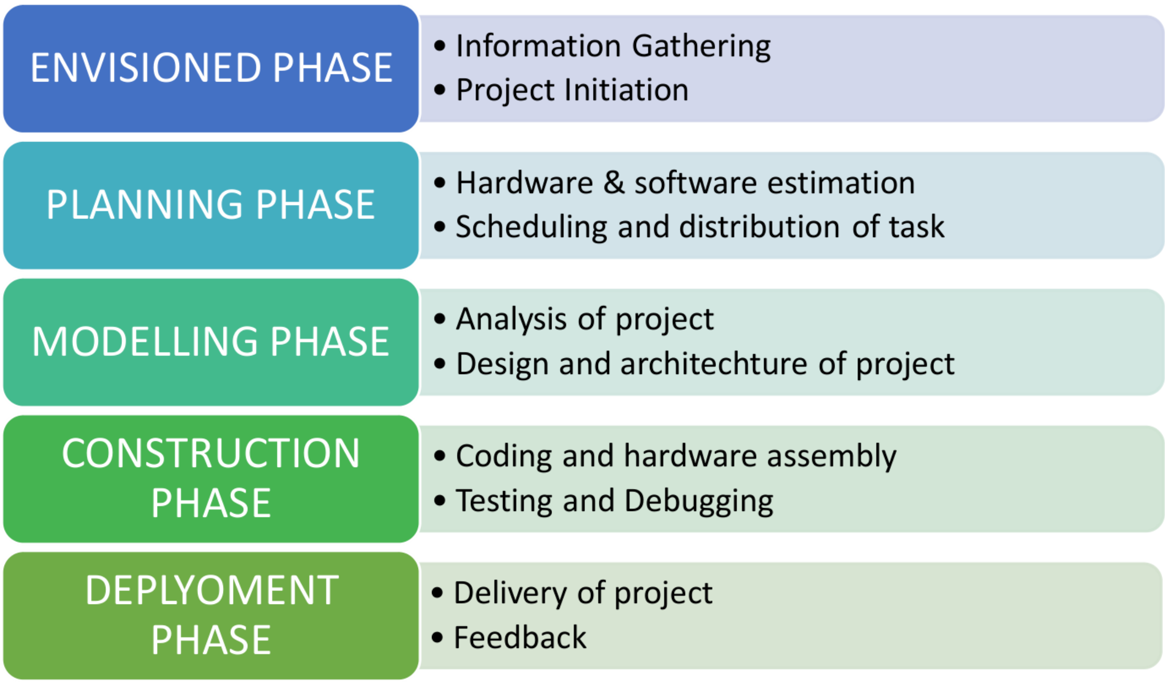

Figure 1. Model of phases in project management.

1.6. Overview and Benefits

The AI-based stethoscope improves diagnostic accuracy, reduces dependency on specialist availability, and allows for early detection of cardiac or respiratory issues. Key benefits include:

- **Accessibility:** Brings diagnostic tools to remote/rural areas.
- **Automation:** Reduces human error in auscultation.
- **Scalability:** Easily replicable and upgradeable.
- **Telemedicine Integration:** Enables remote consultation and monitoring.

1.7. Organization of the Report

The report is organised into the following chapters. Each chapter is unique on its own and is described with necessary theory to comprehend it.

Chapter 2 deals with background survey and review, Chapter 3 has the description of the theoretical aspects that has been acquired to commence the project work.

Chapter 02: Background Review & Survey

2.1. Related Works

Recent advancements in digital health and AI have inspired the development of intelligent stethoscopes that go beyond traditional auscultation. A number of research efforts and commercial products have laid the groundwork for AI-assisted diagnostics in cardiopulmonary healthcare.

One such device is the **Eko DUO**, a digital stethoscope that integrates ECG and heart sound monitoring with a companion mobile app. It enables clinicians to visualize, record, and analyze heart sounds, and it supports AI algorithms to detect murmurs. However, the product's commercial nature limits customization and open-source development.

Another notable work is the **Littmann CORE**, developed in collaboration with Eko, which uses digital amplification and noise cancellation. While it enhances audio clarity, its diagnostic intelligence is relatively limited compared to modern AI techniques.

In academia, several studies have utilized **convolutional neural networks (CNNs)** and **recurrent neural networks (RNNs)** for heart sound classification. For instance, research from the PhysioNet/CinC Challenge 2016 demonstrated that AI models trained on phonocardiogram (PCG) data could classify normal and abnormal heart sounds with performance rivaling experienced cardiologists.

Recent literature also explores the use of **Mel-Frequency Cepstral Coefficients (MFCCs)**, **spectrograms**, and **wavelet transforms** as features for machine learning models. These techniques have proven effective in distinguishing between normal respiration and abnormal sounds like wheezes or crackles.

Despite these advancements, there remains a gap in **low-cost, portable AI-powered stethoscopes** that can be deployed in **resource-limited or remote areas**. This project addresses that gap by leveraging a Raspberry Pi Zero 2 W and open-source tools to build a customizable, real-time diagnostic device suitable for telemedicine.

Chapter 03: Theoretical Aspects

3.1. Internet of Things (IoT)

The **Internet of Things (IoT)** refers to a network of physical objects or "things" that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet.

3.2. Features of IoT

The IoT ecosystem has several distinct **features** that enable its wide applicability across domains like healthcare, smart homes, industry, and transportation. Here are the key features explained:

a) **Intelligence:**

- IoT devices are not just data collectors—they often include **embedded intelligence** through firmware or machine learning.
- Devices can **analyze data**, **make decisions**, or **trigger actions** automatically.
- Example: A smart stethoscope analyzing heart sounds in real-time to detect anomalies.

b) **Connectivity:**

- Core to IoT is **interconnectivity**—devices communicate with each other, cloud platforms, or users.
- Utilizes multiple communication protocols: Wi-Fi, Bluetooth, Zigbee, LoRa, 4G/5G, etc.
- Example: Sensors sending data to a mobile app or cloud server for analysis and storage.

c) **Dynamic Nature:**

- IoT systems are **adaptive** and respond in real-time to changing environments or data inputs.
- Devices can **join or leave networks** dynamically and change behavior based on context.
- Example: A smart thermostat adjusting room temperature based on real-time occupancy.

d) **Enormous Scale:**

- IoT systems are expected to scale to **billions of devices** globally.

- Requires robust infrastructure for **device management**, **data handling**, and **scalability**.
- Example: Smart agriculture using thousands of sensors across large farms.

e) Sensing:

- Sensing is fundamental to IoT. Devices can sense **temperature, pressure, motion, sound,** etc.
- Converts real-world phenomena into digital signals for monitoring and control.
- Example: A microphone in a smart stethoscope capturing heart and lung sounds.

f) Heterogeneity:

- IoT involves a wide variety of **devices, platforms, operating systems,** and **communication protocols**.
- Devices may differ in hardware, data formats, or interfaces.
- Interoperability is essential for a unified system.
- Example: Integrating a Raspberry Pi-based stethoscope with Android apps and cloud servers.

g) Security:

- Security is critical due to the **sensitive data** and widespread connectivity.
- Challenges include **data encryption, authentication, access control,** and **firmware updates**.
- Threats: Hacking, data leaks, unauthorized access.
- Example: Securing medical data collected by wearable IoT devices.

3.3. Advantages of IoT

IoT offers several key **advantages** that enhance productivity, automation, and data-driven decision-making across industries. Here's a breakdown of the major benefits:

a) Communication:

- IoT enables **seamless communication** between devices, systems, and people.
- Machine-to-Machine (M2M) communication improves system coordination.
- Example: A smart stethoscope transmitting real-time data to a doctor’s phone or hospital database

b) **Automation and Control:**

- Reduces the need for manual intervention through **automatic control** based on sensor input.
- Enables real-time responses, improved accuracy, and reduced human errors.
- Example: Automated irrigation systems turning on when soil moisture is low.

c) **Information:**

- IoT collects vast amounts of **real-time data** from the environment.
- This data helps in **analysis, prediction**, and **informed decision-making**.
- Example: Continuous monitoring of a patient's vitals to detect early signs of illness.

d) **Monitoring:**

- Continuous **remote monitoring** of assets, people, and environments.
- Useful for preventive maintenance, health tracking, environmental monitoring, etc.
- Example: Monitoring air quality or heart rate using IoT sensors.

e) **Efficiency:**

- IoT improves **operational efficiency** by automating tasks, optimizing resource usage, and reducing waste.
- Leads to **cost savings** and improved system performance.
- Example: Smart energy meters reducing electricity consumption through usage analysis.

3.4. Disadvantages of IoT

While IoT brings many advantages, it also introduces certain **challenges and disadvantages** that need careful attention, especially in critical domains like healthcare and infrastructure.

a) **Compatibility:**

- ➢ **Lack of standardization** across devices, platforms, and communication protocols can lead to integration issues.

- Devices from different manufacturers may not work together seamlessly.
- Example: A heart monitoring device may not be compatible with certain hospital data systems.

b) Complexity:

- IoT systems involve many components: hardware, software, networks, and cloud services.
- **Designing, deploying, and maintaining** such systems can be complex and resource-intensive.
- Debugging issues across distributed devices adds to the difficulty.
- Example: Maintaining firmware updates for thousands of devices in a smart city

c) Privacy/Security:

- IoT devices often collect **sensitive personal data** (e.g., health, location, behavior).
- Devices are vulnerable to **hacking, data breaches,** and **unauthorized access** if not secured properly.
- Weak encryption, default passwords, and insecure firmware are common threats.
- Example: A medical IoT device being hacked and leaking patient data

d) Safety:

- Malfunctioning IoT devices in critical areas (e.g., healthcare, transportation) can pose **safety risks**.

- Automated systems may cause harm if they behave unexpectedly or are tampered with.
- Example: A smart insulin pump delivering incorrect dosage due to a software bug.

3.5. Application areas of IoT

IoT is transforming numerous industries by enabling smarter decision-making, automation, and real-time data monitoring. Below are the **major application areas** of IoT:

1. Healthcare

- **Remote patient monitoring**, wearable health devices, smart stethoscopes.
- Real-time health tracking and emergency alerts.
- Example: AI-powered stethoscopes detecting abnormal heart or lung sounds remotely.

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2. Smart Homes

- Automation of lighting, appliances, security systems, and HVAC.
- Enhanced comfort, energy savings, and remote control.
- Example: Smart thermostats adjusting temperature based on occupancy.

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3. Industrial IoT (IIoT) / Manufacturing

- Predictive maintenance, asset tracking, and smart factories.
- Real-time monitoring of equipment health and operations.
- Example: Sensors on machines detecting faults before breakdown.

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4. Agriculture

- Precision farming, soil moisture monitoring, automated irrigation.
- Improved crop yield and resource efficiency.
- Example: IoT sensors triggering irrigation based on soil conditions.

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5. Smart Cities

- Intelligent traffic systems, waste management, pollution monitoring.
- Improves urban infrastructure and citizen quality of life.
- Example: Smart streetlights adjusting brightness based on pedestrian presence.

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6. Transportation and Logistics

- Fleet tracking, route optimization, real-time vehicle diagnostics.
- Enhanced supply chain visibility and efficiency.
- Example: IoT devices monitoring cargo conditions during transit.

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7. Energy Management

- Smart grids, automated meter reading, energy usage analytics.
- Reduces energy consumption and facilitates renewable integration.
- Example: Smart meters providing real-time usage feedback to consumers.

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8. Retail

- Smart shelves, customer behavior analytics, inventory tracking.
- Enhances customer experience and operational efficiency.
- Example: Sensors notifying staff when a product is running low.

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9. Environmental Monitoring

- Air quality sensors, weather stations, disaster detection.
- Real-time data collection for environmental research and safety.
- Example: IoT devices alerting authorities about rising flood levels.

3.6. IOT Technologies and Protocols

IoT devices communicate using various **wireless technologies and protocols**, each with its own strengths in range, power consumption, data rate, and use case. Here's an overview of the most commonly used ones:

a) **Bluetooth:**

- **Type:** Short-range wireless communication.
- **Range:** ~10 meters (up to 100m with Bluetooth 5).
- **Data Rate:** Up to 2 Mbps.
- **Power Consumption:** Low.
- **Use Cases:** Wearable devices, smartwatches, wireless earphones, health monitors.
- **Example:** Smart stethoscope transmitting data to a nearby mobile app.

b) **Zigbee:**

- **Type:** Low-power, low-data-rate mesh protocol.
- **Range:** ~10–100 meters.
- **Data Rate:** Up to 250 kbps.
- **Power Consumption:** Very low.
- **Use Cases:** Home automation, smart lighting, industrial IoT.
- **Example:** Sensors in a smart home communicating with a central hub.

c) **Z-Wave:**

- **Type:** Wireless mesh network similar to Zigbee.
- **Range:** ~30–100 meters.
- **Data Rate:** ~100 kbps.
- **Power Consumption:** Very low.
- **Use Cases:** Home automation, smart locks, security systems.
- **Note:** Primarily used in smart home ecosystems.

d) **Wi-Fi**:

- **Type:** High-speed wireless networking.
- **Range:** ~50–100 meters.
- **Data Rate:** Hundreds of Mbps.
- **Power Consumption:** High.
- **Use Cases:** Smart cameras, IoT gateways, home appliances.

- **Example:** Raspberry Pi-based IoT devices using Wi-Fi to upload data to the cloud.

e) Cellular:

- **Type:** Long-range communication via mobile networks.
- **Range:** Very wide (nationwide/global).
- **Data Rate:** Varies by generation (5G offers >1 Gbps).
- **Power Consumption:** High.
- **Use Cases:** Vehicle tracking, remote health monitoring, large-scale IoT.
- **Example:** Remote patient monitoring in rural areas via 4G/5G.

f) NFC(Near Field Communication):

- **Type:** Very short-range communication.
- **Range:** <10 cm.
- **Data Rate:** Up to 424 kbps.
- **Power Consumption:** Very low.
- **Use Cases:** Contactless payments, access control, tag reading.
- **Example:** Scanning a medical device's tag with a smartphone for instant configuration.

g) LoRaWAN(Long Range Wide Area Network):

- **Type:** Low-power, long-range wireless communication.
- **Range:** Up to 10–15 km in rural areas.
- **Data Rate:** 0.3–50 kbps.
- **Power Consumption:** Ultra-low.
- **Use Cases:** Smart agriculture, environmental monitoring, asset tracking.
- **Example:** Soil moisture sensors in large farms reporting data over LoRa.

3.7. Project Layout

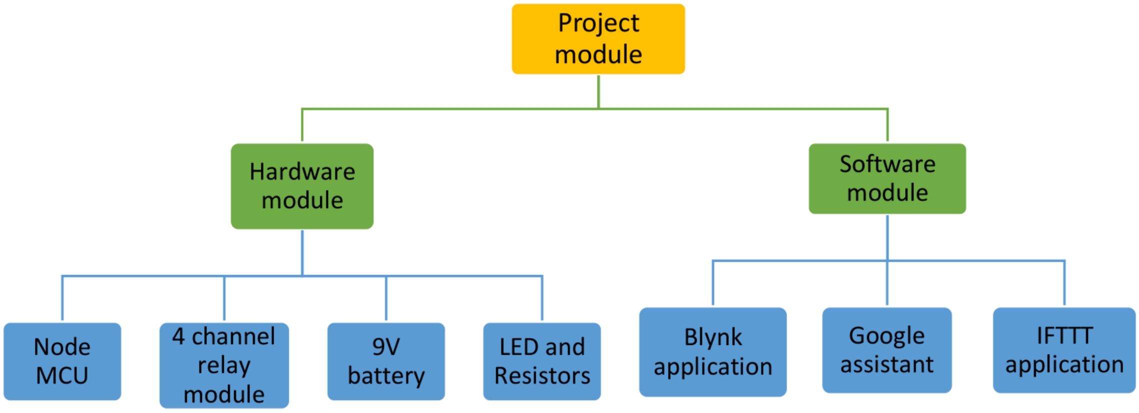


Figure 2. Layout of project module

3.7.1.Brief Description

- The project involves building an **AI-powered smart stethoscope** using IoT and embedded systems.
- Uses a **modified traditional stethoscope** integrated with a **condenser microphone** to capture heart and lung sounds.
- Audio signals are digitized using a **USB sound card** connected to a **Raspberry Pi Zero 2 W**.
- **Signal processing techniques** are applied, including:
- Noise filtering
- Amplitude normalization
- Feature extraction (e.g., MFCCs, spectrograms)
- A **Convolutional Neural Network (CNN)** classifies sounds into categories:
- Normal or abnormal heart sounds (e.g., murmurs, arrhythmias)
- Normal or abnormal lung sounds (e.g., wheezing, crackles)
- Results are displayed locally or sent via **Wi-Fi** to a mobile app or cloud dashboard.
- Designed to support **telemedicine** and **remote healthcare diagnostics**.
- Device is **low-cost**, **portable**, and ideal for **rural or resource-constrained areas**.

Chapter 04: Hardware Requirements

4.1. Raspberry Pi:

The **Raspberry Pi** is a small, affordable, single-board computer developed by the **Raspberry Pi Foundation** to promote computer science education and low-cost hardware development. It has since become widely used in **IoT**, **robotics**, **DIY electronics**, and **embedded systems** projects due to its flexibility, community support, and low power consumption.

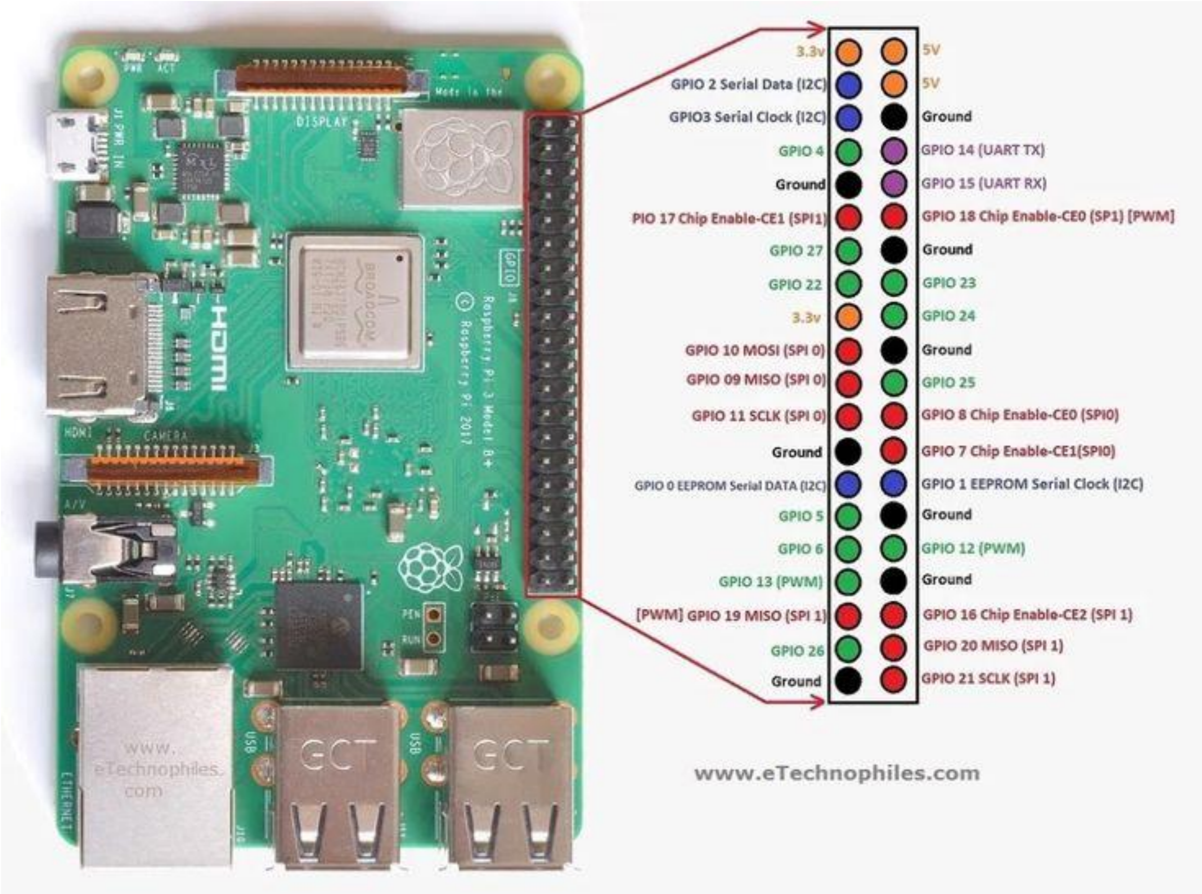
4.1.1. Features:

- **Processor:** ARM-based CPUs (e.g., quad-core Cortex-A53 in Pi 3, Cortex-A72 in Pi 4).
- **Memory:** Varies from 512MB to 8GB RAM depending on the model.
- **Ports:** USB, HDMI, GPIO pins, audio jack, camera interface, display interface.
- **Storage:** microSD card (used as boot drive and storage).
- **Connectivity:** Ethernet, Wi-Fi, Bluetooth (depending on model).
- **Operating System:** Primarily runs **Raspberry Pi OS** (Linux-based), but also supports Ubuntu, Windows IoT Core, etc.

4.1.2. Pin Configuration:

The Raspberry Pi (most models including Pi 3, 4, Zero 2 W) uses a **40-pin GPIO (General Purpose Input/Output)** header for hardware interfacing. These pins include:

Type	Count	Details
Power Pins	4	2× 5V (Pins 2, 4), 2× 3.3V (Pins 1, 17)
Ground (GND)	8	Ground pins (e.g., Pins 6, 9, 14, etc.)
GPIO Pins	26	General I/O for controlling sensors, LEDs, etc.
I2C Interface	2	SDA (Pin 3, GPIO 2), SCL (Pin 5, GPIO 3)
SPI Interface	5	MOSI, MISO, SCLK, CE0, CE1 (Pins 19–24)
UART Interface	2	TXD (Pin 8, GPIO 14), RXD (Pin 10, GPIO 15)



4.2. Stethoscope

***Function:** Captures body sounds such as heartbeats and lung sounds.

***Description:** A traditional acoustic stethoscope modified to include a microphone (in this case, the Boya Mic) at the chest piece.

***Role in System:**

*Transfers the body sound vibrations to the microphone for digitization and analysis.

* Ensures proper contact and sound capture from the patient.

4.3. Boya Microphone

***Function**: Captures body sounds such as heartbeats and lung sounds.

***Description**: A traditional acoustic stethoscope modified to include a microphone (in this case, the Boya Mic) at the chest piece.

***Role in System:**

*Transfers the body sound vibrations to the microphone for digitization and analysis.

*Ensures proper contact and sound capture from the patient.

4.4. LiPo Battery

***Function:**Provides portable power to the Raspberry Pi and other components.

***Description:** A lightweight Lithium Polymer battery offering high energy density and rechargeable capability.

***Typical Specs:**

* Voltage: 3.7V

* Capacity: 1000mAh or more (depending on system power demand)

***Benefits:**

*Rechargeable and compact, ideal for wearable or handheld devices.

* Supports several hours of operation on a single charge.

4.5. DC-DC Charge Discharge Integrated Module

***Function:** Manages charging and voltage regulation from the LiPo battery to the Raspberry Pi.

***Description:** A power management board that allows:

* Safe charging of the LiPo battery via micro-USB or Type-C input.

* Voltage step-up/step-down to maintain a stable 5V output required by Raspberry Pi.

***Integrated Features:**

* Overcharge/discharge protection

* LED indicators for charge status

* Compact and efficient design

4.6. USB Sound Card

*Function: Enables high-quality audio input and output through USB.

***Description:** An external USB audio interface that converts analog audio signals from the Boya mic to digital signals.

***Use Case:**

* Raspberry Pi Zero 2 W lacks a native 3.5mm mic-in port, so this component bridges the gap.
* Ensures clear digital audio capture for processing by AI algorithms.

4.7. Micro USB Type B Mic Patch Straight Plug Adapter Plate Welding Head Breakout Board

***Function:** Provides easy interfacing and soldering for the mic connections to the USB sound card or Raspberry Pi.

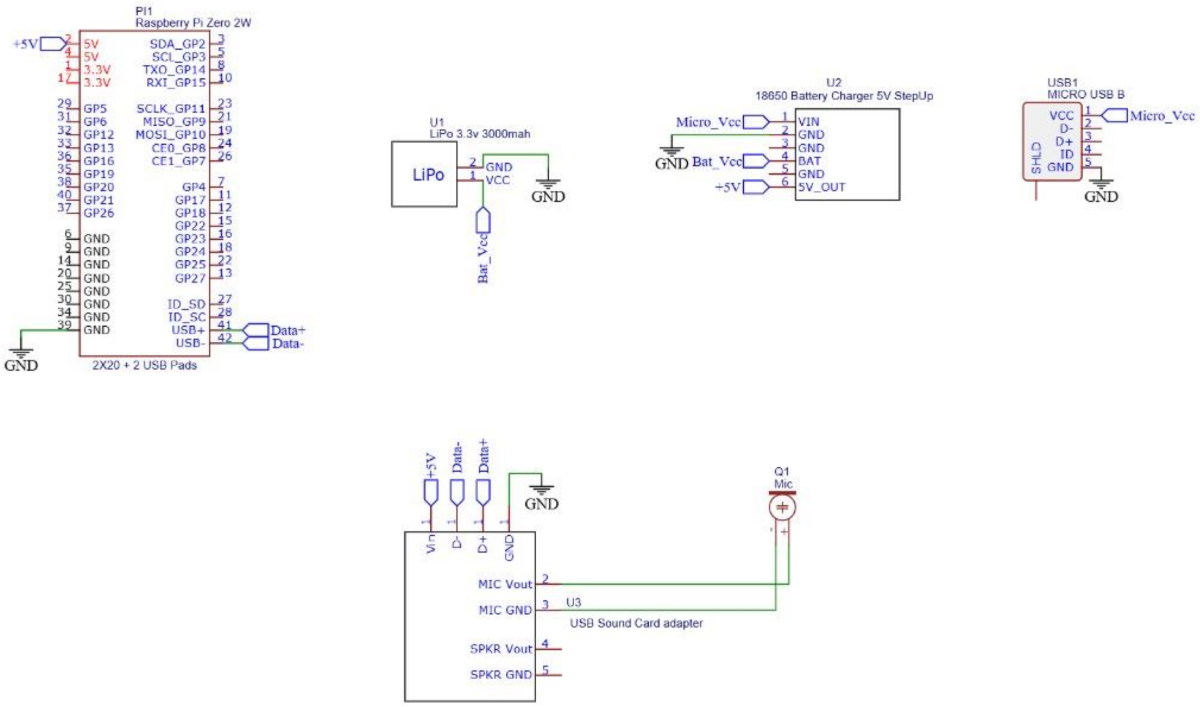
***Description:**

* A small breakout board that converts micro USB Type B connections into solderable pins.
* Supports integration of non-standard or custom-made mic setups.

***Importance:**

* Enables precise wiring in compact devices.
* Ensures secure connections in DIY and prototype hardware builds.

4.8. Circuit Diagram



4.8.1. Components Required:

Table 1. Component listing.

Sl. No.	Component and Specification	Quantity
1.	Raspberry Pi Zero 2W	1
2.	A Stethoscope	1
3.	18W Power Adapter with cable	1
4.	Card Reader	1
5.	5V 3A USB to micro USB cable with ON/OFF switch power control	1
6.	32 GB SD Card	1
7.	Boya Mic	1
8.	Sound Card	1
9.	Micro USB breakout module	1

Chapter 05: Software Requirements

To operate the AI stethoscope effectively, a stable and lightweight software stack is essential. The project leverages the following key software components:

5.1. Raspberry Pi OS (32-bit) – Bullseye Version:

Function: Serves as the operating system for the Raspberry Pi Zero 2 W.

Description: Raspberry Pi OS (formerly known as Raspbian) is a Debian-based Linux distribution optimized for Raspberry Pi hardware.

Bullseye Version Features:

- ✓ Based on Debian 11 (Bullseye), offering enhanced stability, updated packages, and better hardware compatibility.
- ✓ Optimized for 32-bit ARM architecture, ensuring efficient performance on the Raspberry Pi Zero 2 W with its 512MB RAM.
- ✓ Lightweight desktop environment* that can run even on low-resource boards.

Why It's Used?

Ensures maximum compatibility with Raspberry Pi hardware.

Pre-installed tools and libraries like Python, ALSA (Advanced Linux Sound Architecture), and GPIO support streamline development.

Secure and actively supported by the Raspberry Pi Foundation.

5.2. Python Programming Language:

Function: Used for writing the main software code for sound recording, signal processing, and AI model execution.

Description: Python is a high-level, interpreted programming language known for its readability, wide community support, and vast library ecosystem.

Role in the Project:

Audio Capture: Python libraries such as pyaudio or sounddevice are used to capture sound input from the USB microphone.

Signal Processing: Libraries like numpy, scipy, and librosa help in filtering, analyzing, and transforming sound waves (e.g., converting to spectrograms).

Machine Learning: If AI is implemented, frameworks like TensorFlow, PyTorch, or scikit-learn can be used for classification or anomaly detection.

User Interface (Optional): GUI tools such as tkinter or PyQt can be used to create simple user dashboards

.

Why Python?

* Readable syntax and rapid prototyping.

* Abundance of libraries tailored to signal processing and machine learning.

* Seamless integration with Raspberry Pi GPIOs for extended hardware interaction if needed

Tools/Library	**Purpose**
Alsamixer	Adjust and manage audio levels on Raspberry Pi.
Pyaudio	Stream audio from microphone to Python.
Librosa	Analyze and visualize sound signals.
Matplotlib	Plot waveforms,spectrograms,etc.
Scikit-learn	Implement ML algorithm(eg:SVM,KNN).
OpenCV	If video or visual output is required.

5.3. Logic and Flowchart:

The AI stethoscope combines *hardware for audio acquisition* and *software for AI-based analysis*. Here's the end-to-end workflow:

1.Power Initialization:

* The LiPo Battery (3.7V) provides portable power.

* This passes through a DC-DC Boost Converter, which converts it to a stable 5V output.

* Raspberry Pi Zero 2 W uses this 5V to boot up and power peripherals like the USB sound card.

2.Heart/Lung Sound Capture:

* A traditional stethoscope chestpiece is modified to house a Boya condenser mic (Q1).

* The mic picks up low-frequency body sounds (typically 20–200 Hz for heart sounds, and up to 1000 Hz for lung sounds).

3.Analog to Digital Conversion:

* The analog mic signal enters the USB sound card adapter, which digitizes the sound.

* This USB sound card is connected to the Pi's USB data pads (D+, D−), enabling digital audio input.

4.Signal Processing in Raspberry Pi:

* On boot, Python code running in Raspberry Pi OS (Bullseye) handles:

- ❖ Audio acquisition using libraries like sounddevice or pyaudio.
- ❖ Noise filtering and normalization (removing ambient noise).

* **Feature extraction, such as:**

- ❖ Spectrograms (time-frequency analysis)
- ❖ MFCCs (Mel-Frequency Cepstral Coefficients)

5.AI-Based Classification:

* Extracted features are fed into a pre-trained AI model (e.g., CNN, SVM).

* **Model outputs may include:**

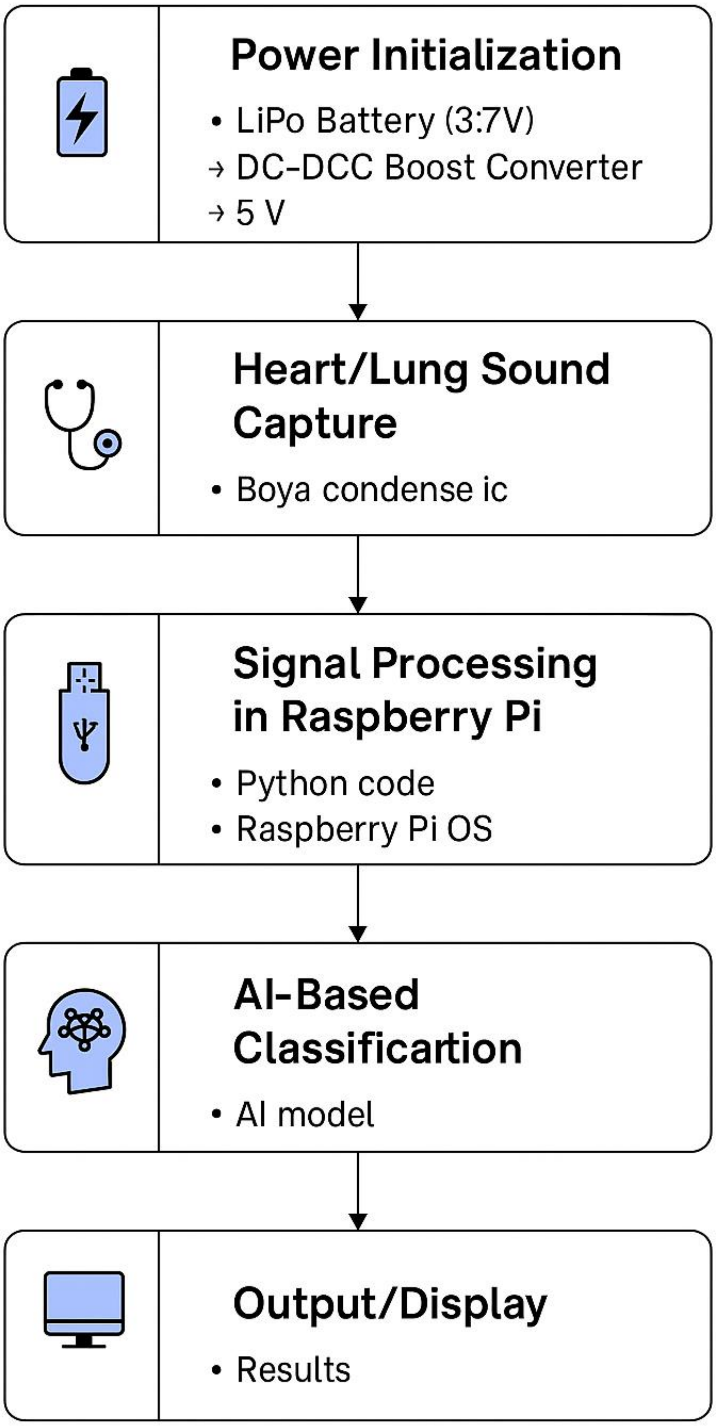
- ❖ Normal heartbeat/lung sound
- ❖ Murmur or arrhythmia (for heart)
- ❖ Wheeze or crackles (for lungs)

6.Output/Display:

*** Results can be:**

- ❖ Shown on a connected screen
- ❖ Sent wirelessly via Wi-Fi
- ❖ Output as an audio alert or visual indicator (e.g., LEDs or GUI)

Flow chart of the Software Implementation



Chapter 06: Project development & Testing Aspects

6.1. Project Development Stages:

The AI stethoscope project was developed in a modular and iterative approach, covering both hardware integration and AI model development.

A. Problem Identification:

* There is a growing need for portable, affordable, and intelligent diagnostic tools, especially in rural or resource-limited areas.
* Traditional stethoscopes rely on a doctor's auditory skill alone, making diagnosis subjective.
* The aim was to digitally capture body sounds and use AI to aid clinical decision-making.

B. Hardware Development:

* **Assembled the physical prototype using:**
    - ❖ Raspberry Pi Zero 2 W (for computation)
    - ❖ Boya mic embedded in a traditional stethoscope
    - ❖ LiPo battery with charging circuitry
    - ❖ USB sound card for analog-to-digital conversion
* Designed and tested power supply circuits to ensure uninterrupted portable use.
* Created and tested schematic and PCB layout using tools like EasyEDA or KiCad.

C. Software Development:

* Installed Raspberry Pi OS (32-bit Bullseye) on the Raspberry Pi.
* **Developed Python scripts for:**
    - ❖ Audio acquisition using sounddevice or PyAudio
    - ❖ Real-time waveform plotting for testing
    - ❖ Preprocessing: band-pass filtering (20–1000 Hz)
    - ❖ Feature extraction: MFCC, spectrograms

*** AI Model Training:**

- ❖ Collected sample audio data (heart/lung sounds)
- ❖ Preprocessed and labeled the data
- ❖ Trained a CNN-based model using TensorFlow or scikit-learn
- ❖ Saved the model and used it for real-time inference on the Raspberry Pi

6.2. Testing Aspects:

A. Unit Testing:

* Each component (battery, mic, USB card, Pi) was tested individually for functionality.

*** Verified:**

- ❖ Mic signal strength
- ❖ USB sound card compatibility with Raspberry Pi
- ❖ Power circuit stability under load

B. Integration Testing:

* Tested the complete data flow from audio input → USB adapter → Raspberry Pi → software → output.

*** Ensured that:**

- ❖ Audio data was captured clearly
- ❖ No power dropouts occurred
- ❖ All Python modules executed as expected

C. Model Validation:

* Used a separate test dataset of labeled audio samples.

*** Evaluated the AI model on:**

- ❖ Accuracy
- ❖ Precision, Recall, F1 Score
- ❖ Confusion Matrix

<u>D. Real-World Testing:</u>

* Conducted tests with actual stethoscope placement on volunteers (with consent).

* **Verified:**
  - ❖ Quality of sound captured
  - ❖ Ability of AI to correctly identify normal vs abnormal sounds

---Tools & Libraries Used---

* Hardware Tools: Soldering kit, multimeter, oscilloscope, PCB design software

* **Software & Libraries:**
  - ❖ OS: Raspberry Pi OS Bullseye (32-bit)
  - ❖ Code: Python 3.x
  - ❖ Libraries: pyaudio, numpy, matplotlib, scipy, librosa, tensorflow

###Improvements During Testing###

* Replaced mic shielding to reduce noise
* Adjusted mic gain and USB power filtering
* Fine-tuned the AI model to reduce false positives

Chapter 07: Conclusion & Future Scope

7.1. Result

AI-enabled stethoscopes have shown promising outcomes across clinical trials, prototypes, and commercial applications:

***Improved Diagnostic Accuracy*:** AI models integrated with digital stethoscopes can classify heart and lung sounds more accurately than general practitioners, especially for detecting:

- Heart murmurs
- Arrhythmias
- Pneumonia
- Asthma and COPD

***Noise Filtering*:** AI helps to filter ambient noise, improving auscultation in noisy environments such as emergency rooms.

***Remote Monitoring*:** Enables telemedicine by allowing auscultation from remote locations with real-time data analysis.

*** Data Logging*:** Records and stores auscultation data for follow-ups or second opinions.

*** Integration with EHRs*:** Supports seamless storage and retrieval of patient diagnostics.

7.2. Conclusion

***Clinical Usefulness*:** AI stethoscopes can augment the diagnostic ability of clinicians, especially in resource-limited settings.

***Early Detection*:** Facilitates early detection of cardiovascular and respiratory conditions.

***Standardization*:** Reduces the subjectivity of traditional auscultation.

***Educational Tool*:** Helps in training medical students with real-time feedback and labeled datasets.

7.3. Limitations

<u>Despite significant advancements, several challenges remain:</u>

***Data Bias*:** Training datasets may lack diversity, affecting accuracy across different populations.

***False Positives/Negatives*:** AI can misclassify certain conditions, leading to unnecessary tests or missed diagnoses.

***Interpretability*:** "Black-box" models often lack explainability, reducing trust among clinicians.

***Cost*:** AI-integrated stethoscopes are more expensive than traditional models.

***Regulatory and Ethical Issues*:** Approval from health authorities (e.g., FDA, CE) is complex; also raises concerns about data privacy and consent.

***Dependence on Internet/Software*:** May be limited in rural or low-connectivity areas.

7.4. Further Enhancement and Future Scope

The AI stethoscope continues to evolve. Possible directions include:

***Multimodal Diagnostics*:** Combining AI auscultation with ECG, PPG, and imaging data for more comprehensive diagnostics.

***On-device AI Processing*:** Enhancing edge computing to reduce reliance on cloud infrastructure.

***Personalized Models*:** Adapting AI for patient-specific baselines using longitudinal data.

***Open Datasets*:** Development of standardized and diverse training datasets.

***Wearable Versions*:** Creating continuous monitoring devices for at-risk patients.

***Global Health Applications*:** Deployment in rural and underserved areas through mobile health initiatives.

***Integration with IoT*:** Real-time data sharing between devices for complete patient monitoring systems.

References

Here are direct **reference links** you can use in your project on **AI-Based Stethoscopes**:

7.4.1. □ Research Papers

1. **A Low-Cost AI-Empowered Stethoscope and Lightweight Model for Detecting Cardiac and Respiratory Diseases**
   □ https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10007545
2. **Artificial Intelligence-Based Stethoscope for the Diagnosis of Aortic Stenosis**
   □
   https://www.sciencedirect.com/science/article/abs/pii/S0002934322003898

7.4.2. □ DIY Project

3. **AI Stethoscope with VIAM – Instructables Guide**
   □ https://www.instructables.com/AI-Stethoscope-With-VIAM/

7.4.3. □ Industry Article

4. **IEEE Spectrum: A Smart Stethoscope Puts AI in Medics' Ears**
   □ https://spectrum.ieee.org/a-smart-stethoscope-puts-ai-in-medics-ears

---

Appendix 01

A01.1. Code Listing

```
import sounddevice as sd

import numpy as np

import scipy.signal as signal

import matplotlib.pyplot as plt

from scipy.io.wavfile import write


# === Configuration ===

fs = 16000          # Sampling rate in Hz

duration = 10       # Duration to record (seconds)

lowcut = 20.0       # Bandpass filter low cutoff (Hz)

highcut = 150.0     # Bandpass filter high cutoff (Hz)


# === Step 1: Record Audio ===

print("Recording heartbeat for", duration, "seconds...")

recording = sd.rec(int(duration * fs), samplerate=fs, channels=1,
dtype='float32')

sd.wait()

signal_raw = recording.flatten()
```

```
# Save raw audio
write("raw.wav", fs, recording)
print("Raw audio saved as 'raw.wav'.")

# === Step 2: Bandpass Filter ===
nyq = 0.5 * fs
b, a = signal.butter(4, [lowcut / nyq, highcut / nyq], btype='band')
filtered = signal.filtfilt(b, a, signal_raw)

# Save filtered audio
write("filtered.wav", fs, filtered.reshape(-1, 1))
print("Filtered audio saved as 'filtered.wav'.")

# === Step 3: Peak Detection ===
peaks, _ = signal.find_peaks(filtered, distance=fs * 0.5, prominence=0.01)

# Estimate BPM
num_beats = len(peaks)
bpm = (num_beats / duration) * 60
print(f"Detected {num_beats} heartbeats in {duration} seconds.")
```

```python
print(f"Estimated BPM: {bpm:.2f}")


# === Step 4: Plotting ===
plt.figure(figsize=(12, 4))
plt.plot(filtered, label='Filtered Signal')
plt.plot(peaks, filtered[peaks], 'rx', label='Detected Beats')
plt.title(f"Heartbeat Signal (Estimated BPM: {bpm:.2f})")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.legend()
plt.tight_layout()
plt.savefig("heartbeat_bpm_plot.png")
plt.show()
```

A01.2. Main Code

```
import sounddevice as sd
import numpy as np
import scipy.signal as signal
import matplotlib.pyplot as plt
from scipy.io.wavfile import write

# Parameters
fs = 16000       # Sample rate
duration = 10    # seconds
lowcut = 20.0    # Bandpass low cutoff
highcut = 150.0  # Bandpass high cutoff

print("Recording heartbeat for", duration, "seconds...")
recording = sd.rec(int(duration * fs), samplerate=fs, channels=1, dtype='float32')
sd.wait()
signal_raw = recording.flatten()

# Save raw audio
write("raw.wav", fs, recording)

# Bandpass filter
nyq = 0.5 * fs
b, a = signal.butter(4, [lowcut / nyq, highcut / nyq], btype='band')
filtered = signal.filtfilt(b, a, signal_raw)

# Save filtered audio
write("filtered.wav", fs, filtered.reshape(-1, 1))

# Peak detection (heartbeat detection)
peaks, _ = signal.find_peaks(filtered, distance=fs*0.5, prominence=0.01)

# Estimate BPM
num_beats = len(peaks)
bpm = (num_beats / duration) * 60
print(f"Detected {num_beats} heartbeats in {duration} seconds.")
print(f"Estimated BPM: {bpm:.2f}")

# Plot
plt.figure(figsize=(12, 4))
plt.plot(filtered, label='Filtered Signal')
plt.plot(peaks, filtered[peaks], 'rx', label='Detected Beats')
plt.title(f"Heartbeat Signal (Estimated BPM: {bpm:.2f})")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.legend()
plt.tight_layout()
plt.savefig("heartbeat_bpm_plot.png")
plt.show()
```

A01.3. Libraries

- numpy
- scipy
- matplotlib
- librosa
- sounddevice
- tensorflow
- pyaudio

Appendix 02

A02.1. Project Proposal Form

The project proposal form was prepared and duly signed from our Faculty-in-Charge Dr. Biswaranjan Swain. The same is attached at the last of this report.

A02.2. Project Management

#	Component	Individual Contributions in %			Total
		Abhilash Mishra	**Rohan Kumar sahu**	**Durga Madhaba Dash**	
1.	Planning	33.3	33.3	33.3	100
2.	Background Research and Analysis	33.3	33.3	33.3	100
3.	Hardware design	33.3	33.3	33.3	100
4.	Software design	33.3	33.3	33.3	100
5.	Testing	33.3	33.3	33.3	100
6.	Final Assembling	33.3	33.3	33.3	100
7.	Project report writing	33.3	33.3	33.3	100
8.	Presentation	33.3	33.3	33.3	100
9.	Logistics	33.3	33.3	33.3	100

A02.3. Bill of Material

Table 1. Component listing.

#	Component	Specification	Unit Cost	Quantity	Total
1.	Raspberry Pi Zero 2 W	Main control board	Rs.1560	1	Rs.1560
2.	Micro SD card 32GB	Boot and Storage	Rs.395	1	Rs.395
3.	Micro SD card reader	For flashing OS	Rs.50	1	Rs.50
4.	USB to Micro USB cable with switch	Power Supply	Rs.185	1	Rs.185
5.	Stethoscope	Check the Heart Beats	Rs.350	1	Rs.350
6.	Boya Microphone	Takes input audio	Rs.759	1	Rs.759
7.	1800mAh LiPo Battery	Power supply to the circuit	Rs.839	1	Rs.839
8.	Sound card	Capture Audio	Rs.107	1	Rs.107
9.	Micro USB break out module	Use to access 5V and GND lines micro USB cable or charger	Rs.48	1	Rs.48
10.	3-D model of the package		Rs.200(per hour)	5 hours and 30 minutes	Rs.1100
11.	Potronics 18W Adapter +type c cable	Power adapter	Rs.550	1	Rs.550
Grand Total					Rs.5943

Appendix 03

A03.1. Data Sheets

Here is a **datasheet-style summary** for **heartbeat sound**, which is essential for developing medical devices like an **AI stethoscope**:

7.1. □ Heartbeat Sound – Technical Datasheet

7.1.1. □ 1. Signal Characteristics

Parameter	Description
Frequency Range	20 – 150 Hz
Duration of Heartbeat	~0.8 seconds (for 75 BPM)
Components	S1 (lub), S2 (dub); sometimes S3/S4
Amplitude	Low (typically < 1V peak-to-peak on mic input)
Periodicity	Repeating waveform based on heart rate
Typical Heart Rate	60 – 100 BPM (beats per minute) in healthy adults
Waveform Shape	Impulsive / transient
Energy Concentration	Below 150 Hz (main energy in S1 and S2)

7.1.2. □ 2. Recording Requirements

Parameter	Description
Microphone Type	Condenser mic with flat response < 200 Hz
Sample Rate	≥ 1000 Hz (Recommended: 4000–16000 Hz)

Parameter	Description
Bit Depth	16-bit minimum
Channels	Mono
Recording Environment	Quiet room, minimal ambient vibration
Amplification	Required (especially when using electret mics)

7.1.3. □ 3. Clinical Correlations

Heart Sound Component	Physiological Meaning	Possible Abnormalities
S1 (Lub)	Mitral and tricuspid valve closure	Diminished in heart failure
S2 (Dub)	Aortic and pulmonary valve closure	Split in bundle branch block
Murmur	Turbulent blood flow	Valve defects (stenosis, regurg)
Gallop (S3/S4)	Abnormal heart sounds	Heart failure, hypertrophy

---

7.1.4. □ 4. Use in AI Models

Application	Input Type	Feature Extraction Methods
Classification (Normal vs Murmur)	WAV/PCM audio	MFCCs, spectrograms, envelope detection
Heart Rate Estimation	Filtered waveform	Peak detection, autocorrelation

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